Fixes and findings behind the improved benchmark results

Daytona sandbox benchmarks: HQ Vera, Graviton5, Turin, EPYC 9575F / 9755 and earlier cells

Compiled 2026-10-14 from the three working sessions (Vera / project-wide since 1 Sep, Graviton5, Turin), read up to about 21:00 UTC. The Turin density ceiling and the Vera / 9575F reruns with the fixes on were still running.

How to read this. Section 1 lists platform code changes, section 2 host and configuration fixes, section 3 client and harness fixes. Section 4 lists findings that changed how earlier numbers must be read — some "improvements" were really corrections of a contaminated measurement. Section 5 is what was tested and made no difference; section 6 is still open. Numbers are before → after on the host named.

1. Platform code fixes

1.1. Create path: four fixes, each behind a flag (HQ Vera, 14 Oct)

Fix	What changed	Effect
#3260 `RLP_INITDISK_MODE=nofsync`	No 16 MiB init-disk copy with fsync per sandbox	That step 300–322 ms → **9 ms**
#3261 `RLP_CGROUP_POOL=1400`	cgroups pre-created when the runner starts	188–201 ms → **24 ms** under a burst 454 → 154 ms
#3265 `RLP_NEIGH_PERMANENT=1`	Permanent neighbour entry; removes a 580 ms stall after a sandbox resumes	Stalls 4/60 → 0; single-create p99 657 → 73–81 ms
#3269 `RLP_API_LISTEN_BACKLOG=4096`	API listen queue 128 → 4096	Dropped connections 424–687 → 0; connect p90 1.06 s → 51 ms

All four on — measurement	Fixes off	Fixes on	Change
Vera, own client, idle image, 1k: request → running p50	1.93 s	**1.05 s**	−45 %
Vera, own client: all 1,000 running	2.95–3.88 s	2.33–2.37 s	
Vera, single create p50 / worst	67 / 658 ms	64 / 73 ms	
Vera, Daniel's harness, idle image, 1k: p50	2.07–2.39 s	**1.51–1.62 s**	−30 %
Vera, Daniel's harness: p95	3.30–3.76 s	2.54–2.61 s	−27 %
Graviton5, idle image, 1k: typical sandbox	≈ 2.2 s	**0.85 s**	
Graviton5, idle image: p95	≈ 3.4 s	1.55 s	
Agent image startup, both hosts	10–14 s	unchanged	none
Throughput (Vera 352 / Graviton5 704)	26.7 / 22.8 jobs/s	26.7 / 22.9 jobs/s	none

Remaining burst limits after the fixes: waiting for one of the runner's 256 in-flight slots (421 ms) and Firecracker start (200 ms). Open: on Graviton5, 5 of 3,000 fixes-on sandboxes waited 5.1–5.7 s for their toolbox (one run 7.6 s). These fixes live only in the benchmark build; none is merged.

1.2. Other platform fixes

Fix	Effect	Status
Sudo storm (#1754, `1b2da010a`; bounded privileged-exec pools)	BM1/BM2 at 1,800 sandboxes: 1,720 of 1,880 failed → **0 of 3,600** p50 260 → 20 s. At 1,408: p50 13.8 → 12.8 s	Upstream
#3212 Firecracker socket wait 3 s → 30 s (`RLP_FC_SOCK_WAIT_MS`)	Vera: 177–304 timeouts and 6–10 runner self-cordons per 1k → 0. 9575F: 15–48 → 0. Medians unchanged; removes 50–80 s stragglers	Knob in bench build; issue filed
#1060 EROFS 4 KiB blocks on 64 KiB-page hosts	Vera creates: always failing → working	Bench build; PR open
#1066 balloon off on 64 KiB-page hosts	Guest memory corruption: 14/64 failed → 0/544	Applied on all cells
#768 `RLP_CLOCK_REALTIME=0` (nested KVM)	Snapshot clones restored on DigitalOcean: 10 s → under 1 s	PR
Create-queue cap `CREATE_MAX_QUEUE_SECS_CAP`	Hard-coded to 60 s; the env setting is clamped. It stopped Turin's 5,000 level (962 of 1,059 failures)	Patched locally on Turin
Overload gate `RLP_MAX_QUEUED_CREATES` (default 8,000)	Turin's next stop at 8,000 live; raised to 40,000 → past 8,000 with 0 failures	Turin, running

2. Host and configuration fixes

Fix	Host(s)	Effect
Stopped `rlp-cpu-weight-floor.sh` (busy loop, 10 days)	9575F	1k median 18.6 → 9.7 s; all-ready 36 → 20 s; 500 sandboxes 20.4 → 7.2 s. It ran during the published 2 Oct runs
Hostname added to `/etc/hosts`	RedSwitches, 9575F, all new cells	Every `sudo` waited on DNS: 113 s (or >25 s) → 0.2 s. The runner could not start without it
ARP table `gc_thresh3` 1,024 → 65,536 (v4 + v6)	BM1/BM2, OCI, atl1, all cells	704 sandboxes: 1,324 failures → 0; 1,056: host unreachable → 98 % succeeded; peak 55.9 → 62.7 jobs/s
Per-sandbox memory burst cap 32 GiB → 512 MiB	BM1/BM2	2,112 sandboxes: 14.4 → **137.6 jobs/s** 1,408: 12.3 → 83.2; OOM kills 406 → 0
Client open-files limit 1,024 → 1M, plus HTTP connection pool	Vultr BM	1,056 sandboxes: 484 failures → 0

Toolbox wait 60 → 300 s	BM pair	1,408: 1,210 failures → 1 (later credited mostly to the sudo fix)
Teardown concurrency 128 → 256	BM pair	Teardown no longer trips the live-VM cap between levels
E2B huge pages released (400,000 → 0)	HQ Vera	Usable memory 2 GB → 1.44 TB; a 2k run was impossible before
udev iscsi rule shadowed; libvirtd and fwupd stopped	OCI, HQ Vera, Turin, Graviton5	No background work on every sandbox interface change (OCI alone: 574 → 461 failures, near noise)
Postgres and NATS off the boot disk (#707)	Ashburn, OCI	fsync 1.3 ms → 0.03 ms (47×); 1,079 → 14 µs (75×)
12-disk RAID10 removed (60 h resync)	Turin	Matches Vera's single-NVMe layout; no resync competing with runs
Stale firewall tables removed	HQ Vera	Guest replies had been dropped
Policy-routing rule (priority 95)	atl1	Toolbox unreachable → 0 failures to 704
Capacity for the ceiling test: netns pool 6,200, max live 8,000, /18 subnet	Turin	A /20 subnet caps a host near 4,094 sandboxes
Lower API minimums / memory-per-CPU floor	Vera, RedSwitches	Small shapes had been silently rounded up
Runner prerequisites (#724), EVENTS stream (#708), wedged NATS consumer, pre-warmed template	early cells	Failing creates fixed; a cold template caused a second 60 s delivery
Leaked sandboxes, taps, `/run/fc` and scratch cleaned	BM pair, RedSwitches	Next run clean (70.3 jobs/s at 704)
Pin Vera to one socket for the agent burst	HQ Vera	22 → 10 s — only with CPU-heavy guests; pinning hurts Turin and Graviton5

3. Client and harness fixes

Fix	Effect
`mpfleet.py`, 16 client processes	"Accept" 2.9–5.8 s → 0.12–0.5 s. Turin 1k: 10.2 → 7.9 s. A single Python process was queueing requests
SDK `int(memory)*1024` / `int(disk)*1024` bug	Fractional sizes became 0 → HTTP 400, or a silent 1 GiB default. Every Turin ceiling run reserved 1 GiB scratch and stopped at exactly 6,742; with 64 MiB it went past 8,000
Teardown in `finally` + SIGTERM handler	Interrupted runs no longer leak their fleet (the cause of the lockouts)
`RLP_HTTP_READ_TIMEOUT_S=120`	One wedged sandbox no longer turns a level into 1.47 jobs/s or a 26-minute stall
`--rlp-cpu-max 1` required	Without it the harness sends `mode=dedicated`, which fails with HTTP 400
`.env` pointed at `127.0.0.1` (EPYC 9755)	It pointed at the public API: 100 % failure → 0 at 880 / 1,056
HTTP pool 100 → 600 (Daniel)	Short-job rate on Vera 82 → 129
`bench-preflight`	Refuses to run with the busy loop, leftover sandboxes, a filling netns pool, huge pages, a low ARP limit or the balloon on
Warm-up after snapshots are deleted	Graviton5's "2.3× memory on 64 KiB pages" was full boots; the real figure is +8 %

4. Findings that changed how results read

- **The agent image measures its own workload, not startup.** Each sandbox burns 3–4 CPU-seconds at boot. With an idle image 1,000 sandboxes start in ≈ 4 s on Vera and the 9575F and 3.8 s on Graviton5, not 10–30 s. Daytona's own CPU cost is ≈ 28 ms per sandbox.
- **"Control plane accepts the request" was the single-process client**, not the control plane.
- **Pinning helps only Vera, and only with CPU-heavy guests.** Vera's cross-socket memory costs 3.08× vs 1.25× on Graviton5. Pinning halves throughput everywhere. The "NUMA is the startup lever" claim was withdrawn on #391.
- **Published-data corrections.** The 9575F 2k mean keeps a 151 s outlier and was taken with the busy loop on (clean 76–91 s). The 9755's 88.4 s drops a run (all four average 91.1 s). Vera's 2k throughput point had 53–627 failures. September and October startup figures come from different hosts and builds.
- **A second operator invalidated runs.** Daniel's agent worked on HQ Vera overnight; "API restart makes it fast" was retracted. Rule: one operator per box.
- **Measurement traps.** A 0.5 s poll interval sets small-burst numbers; `checksum_ok` is false on any failure; capacity-gate failures are not platform failures; middle throughput levels vary ±30 %, only the plateau is stable.
- **Throughput = cores × per-core speed when packed.** Platform overhead is 0.5–0.6 % of each job. Clock under full load: 9575F −24 %, Vera −6 %, Graviton5 −3 %. Memory bandwidth is not a factor.
- **64 KiB host pages:** no speed gain on Graviton5 (≤ 1.5 %); ≈ 8 % more host memory per sandbox, enough to OOM one NUMA node at 1,408 sandboxes.

5. Tested with no effect

- Event auditing off
- API restarts
- Quiet-progress runner
- `cpuidle.off` + `init_on_alloc=0`
- KVM `halt_poll_ns`
- `RLP_VM_MAXACK`
- `VM_CONCURRENCY` 192 vs 512
- API DB pool 64 → 256
- Proxy scaling (1 vs 4)
- Dedicated control-plane cores
- NVIDIA SMT Firecracker fork
- `RLP_MIGRATABLE=0` (#3213)
- Client pool / polling / disk size
- Memory bandwidth, kernel version
- cgroup `favordynmods`, XFS prealloc
- Create-queue env (clamped to 60 s)

6. Still open

- **#3213** — 2-vCPU snapshot clones crash ≈ 7 % on Vera; 1-vCPU ≈ 0.02 %.
- **#3229** — a sandbox counts as running as soon as its toolbox port accepts, so a hung guest looks healthy.
- **Graviton5 with fixes on** — rare 5 s toolbox waits (5 in 3,000).
- **Turin idle-image startup** — ≈ 8 s per 1,000 vs ≈ 4 s on Vera and the 9575F; unexplained.
- **Pending runs** — Vera and 9575F reruns with the fixes on; Turin density ceiling above 8,000.
- **Never tested** — `preempt=none` vs `full`; 4 KiB vs 64 KiB host pages on Vera itself.